ModelIt!

Student Activity Pack

Can You Outsmart a Disaster?

How Science Helps Communities Survive Natural Hazards

Grade 6 | NGSS MS-ESS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What natural disasters have you heard about or experienced? What happened?

2 Why do you think some buildings collapse in earthquakes while others don't?

3 If you knew a hurricane was coming in 48 hours, what would you do?

4 Draw a community plan that would be safe from flooding.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Hazard Intensity | Population Density | Warning Time | Building Design

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Unprepared City

Poor buildings, no warning

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Prepared Community

Strong buildings, 48-hour warning

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Monster Storm

Maximum hazard intensity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Emergency Response Time — How quickly first responders can reach affected areas after a disaster strikes

Community Education — How well-informed residents are about hazard risks and evacuation procedures

Infrastructure Age — Older buildings and infrastructure are more vulnerable; newer construction follows stricter codes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some communities survive natural disasters better than others — what makes the difference between tragedy and resilience? Use evidence from your simulation to support your answers.

1 Why did the 2010 Haiti earthquake kill 230,000 people while a stronger earthquake in Japan killed far fewer?

2 Is it possible to make a community completely disaster-proof? Why or why not?

3 How does population density change the impact of a natural hazard?

4 Should governments force people to move away from high-risk areas? What are the ethical considerations?

5 How does your model help decision-makers spend limited money on the most effective protections?

8 STEM Challenge: Design a Disaster-Resilient Community

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YOUR ENGINEERING MISSION

Design a community plan for a region that faces multiple natural hazards (earthquakes AND flooding) that minimizes casualties and property damage within a limited budget.

THE SCENARIO

A coastal city near a fault line needs a comprehensive disaster resilience plan. Your urban planning team has a $10 million budget and must decide how to allocate resources among warning systems, building upgrades, evacuation routes, and emergency shelters. Use your model evidence to justify every budget decision.

Engineering Design Thinking

1. HOW WILL YOU DIVIDE THE $10 MILLION BUDGET AMONG DIFFERENT RESILIENCE STRATEGIES?
2. WHICH AREAS OF THE CITY SHOULD BE RESTRICTED FROM RESIDENTIAL DEVELOPMENT?
3. WHAT BUILDING CODES WILL YOU REQUIRE FOR NEW CONSTRUCTION?
4. HOW WILL YOU ENSURE THAT WARNING SYSTEMS REACH EVERY RESIDENT, INCLUDING THOSE WITHOUT SMARTPHONES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Emergency Management Specialists and Civil Engineers design systems that protect communities from natural hazards. They plan evacuations, design resilient infrastructure, and save lives. They earn $60,000–$120,000/year.